

$$m(x) = \frac{1}{n} \sum_{i=1}^n x_i$$

Q 1.1

$$m(a+bX) = \frac{1}{n} \sum_{i=1}^n (a+bX)$$

$$= \frac{1}{n} \left(\sum_{i=1}^n a + \sum_{i=1}^n bX \right)$$

$$= \frac{1}{n} \left(Na + b \sum_{i=1}^n X \right)$$

$$= \cancel{\frac{Na}{n}} + b \cdot \frac{1}{n} \sum_{i=1}^n X$$

$$= a + b \cdot \frac{1}{n} \sum_{i=1}^n X$$

$$= a + b \cdot m(X)$$

Q2.2

$$\text{cov}(X, a + bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))((a + by_i) - m(a + bY))$$

$$m(a + bY) = a + b \cdot m(Y)$$

$$a + by_i - a - b \cdot m(Y) = by_i - b \cdot m(Y) \\ b(y_i - m(Y))$$

$$\text{cov}(X, a + bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) \cdot b(y_i - m(Y))$$

$$= b \cdot \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(y_i - m(Y))$$

$$= b \cdot \text{cov}(X, Y)$$

$$\therefore \text{cov}(X, a + bY) = b \cdot \text{cov}(X, Y)$$

Question 1.3

$$\begin{aligned}\text{cov}(a + bX, a + bX) &= b^2 \text{cov}(X, X) \\ &= \frac{1}{N} \sum_{i=1}^N ((a + bx_i) - m(a + bX))((a + bx_i) - m(a + bX))\end{aligned}$$

From prior question)

$$(a + bx_i - a - b \cdot m(X)) = b(x_i - m(X))$$

$$= \frac{1}{N} \sum_{i=1}^N (b(x_i - m(X)))(b(x_i - m(X)))$$

$$= \frac{1}{N} \sum_{i=1}^N b^2 (x_i - m(X))^2$$

$$= b^2 \cdot \left(\frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2 \right)$$

$$= b^2 \cdot \text{cov}(X, X)$$

$\therefore \text{cov}(a + bX, a + bX) = b^2 \text{cov}(X, X)$
is true

$$\text{cov}(X, X) = s^2$$

$$\text{cov}(X, X) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(x_i - m(X))$$

$$= \frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2$$

Knowing that $s^2 = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2$,

$$\boxed{\text{cov}(X, X) = s^2}$$

Let $g(x)$ be decreasing, $g(x) = x^2$ where $x \geq 0$
let $N = 3$, $x_1 = 1$, $x_2 = 2$, $x_3 = 3$

$$m(g(x)) = \frac{1}{3} \sum_{i=1}^3 x_i^2 = \frac{1^2 + 2^2 + 3^2}{3} = \frac{14}{3}$$

$$g(m(x)) = \left(\frac{1}{3} \sum_{i=1}^3 x_i \right)^2 = \left(\frac{1+2+3}{3} \right)^2 = 2^2 = 4$$

$m(g(x)) \neq g(m(x))$ will always be true

No, this might only be true for when $g(x)$ is a straight line function. The example above proves that the statement is wrong.